Invoice and PO review packets

The system in one sentence

Explain invoice differences and give an AP analyst a clear review packet.

Key design calls

Start with review drafts. Rejected: Automatic action. Why: Staff retain every decision.

Use supplied sources. Rejected: Open browsing. Why: Claims need checked evidence.

Code checks hard facts. Rejected: Model authority. Why: Values and access must be repeatable.

Who acts and who decides

Action	Autonomy
Check supplied evidence	acts alone
Draft an explanation	suggests
Send or take action	human decides

Top risks by RPN

FM4 | Revoked source access | RPN 108

FM1 | Currency differs | RPN 96

FM2 | Ambiguous OCR digit | RPN 84

FM3 | Wrong supplier PO | RPN 84

FM6 | Stale concurrent replacement | RPN 84

Launch blockers and owners

H1, H3, H5: Read supplied invoices and read-only...; One workday response target, coordinator...; Proposed quality target is 95%... Owner: AP operations lead.

H2: Authorized staff supply source bundles... Owner: Integration owner.

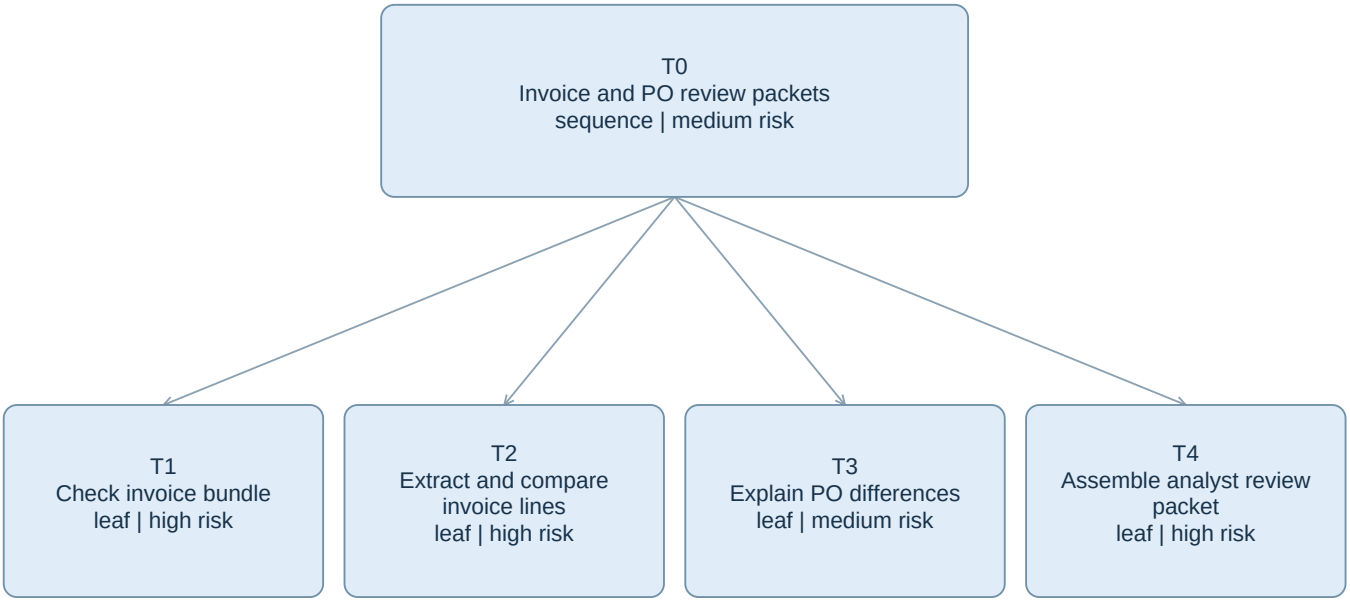
Illustrative cost per case

USD 0.00720 per case in the model budget. This is an estimate, not a current provider quote. Staff and tools cost extra.

Review status: self-review; round 3; pass.

Proposed design. Full rules and blockers: design.json and assumptions.md.

Task tree



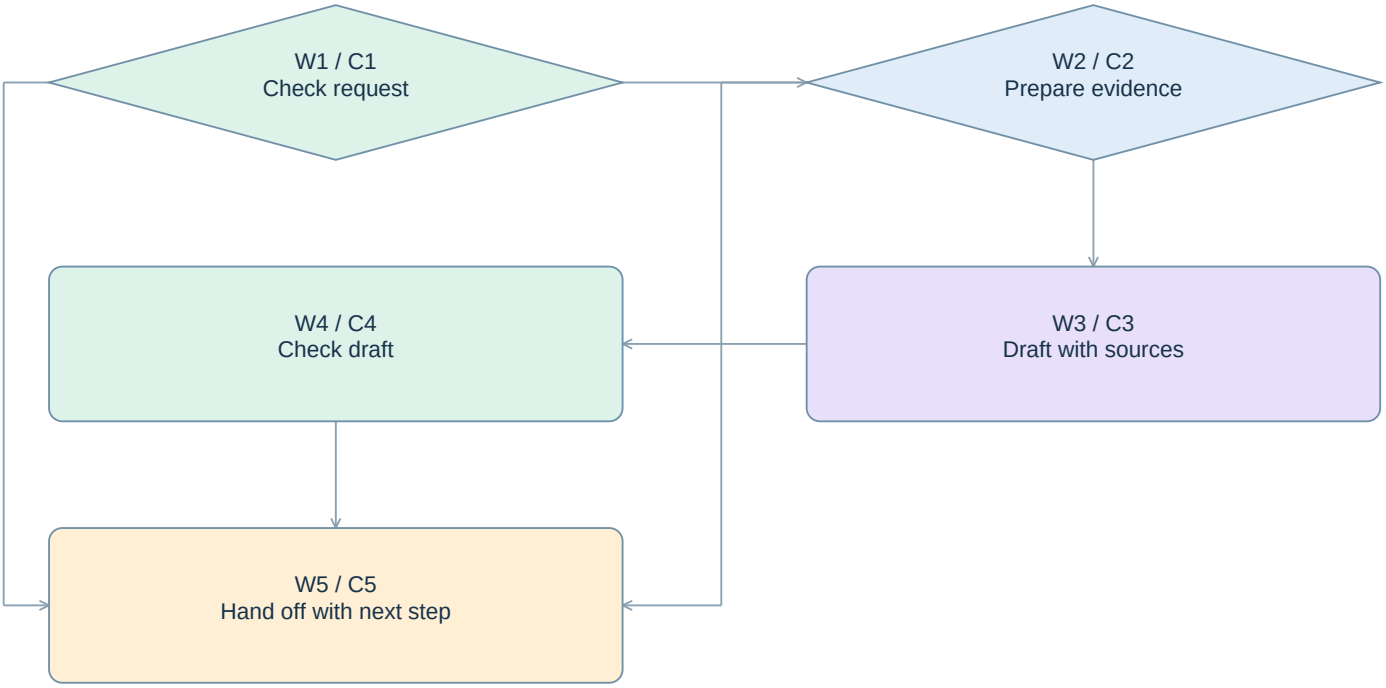
Labels show each leaf mode and risk. Full task inputs, examples and limits are in design.json and PRD.md.

Plain words and scope

Read supplied invoices and read-only PO snapshots. Explain line differences for analyst review. Exclude posting, payment, supplier changes, tax advice and approval of invoices.

A fingerprint is a check that the approved details have not changed. A record ID is proof from a tool that the change was saved.

Workflow diagram



Diamonds mark choices. Arrows show all routes, including retry and recovery. Exact branch rules are in architecture.md.

Colour	Role
#E8E0FA	Agents (LLM)
#E0EDF8	Code: tools
#DDF2E9	Code: validation
#FFF0D5	Code: human triggers

Top risks and release checks

Risk heat table - severity x occurrence

Each cell names the risks at that severity and occurrence. The full scores, causes, controls and owners are in fmea-register.csv.

S / O	1	2	3	4	5	6	7	8	9	10
10	-	-	-	-	-	-	-	-	-	-
9	-	-	FM4	-	-	-	-	-	-	-
8	-	-	FM1	-	-	-	-	-	-	-
7	-	-	FM2, FM3, FM6, FM7, FM9, FM10, FM11, FM12, FM13	-	-	-	-	-	-	-
6	-	-	FM5, FM8	-	-	-	-	-	-	-
5	-	-	-	-	-	-	-	-	-	-
4	-	-	-	-	-	-	-	-	-	-
3	-	-	-	-	-	-	-	-	-	-
2	-	-	-	-	-	-	-	-	-	-
1	-	-	-	-	-	-	-	-	-	-

Risk	Control summary / full register linked
FM1 Currency differs	Code prohibits cross-currency comparisons; analyst obtains a correct PO.
FM2 Ambiguous OCR digit	No guessed numeric fields; null extraction blocks computed comparison.
FM3 Wrong supplier PO	Trusted scope matches invoice and PO before document contents reach model.
FM4 Revoked source access	Fresh access checks before source reads and final ready output.
FM5 Model output unavailable	Bounded call and explicit fallback; no ready success claim.
FM6 Stale concurrent replacement	Compare revision at intake and handoff; same requestId/revision returns same artifact; changed intent needs next revision.
FM7 Missing receipt	C4 verifies receipt before ready. A failed receipt remains unconfirmed; staff may create a new local export. No external commit exists.
FM8 Cancellation	Cancellation precedes draft; cancellation during a call discards response and returns cancelled receipt.
FM9 Overdue human review	Packet includes proposed one-workday target, owner and backup. An operator checks pending list at daily opening; overdue remains visibly pending. Receiving...
FM10 Exact duplicate request	Return same local artifact for exact request/version; changed revision never reuses a draft.
FM11 Conflicting supplied evidence	Do not pick one conflicting record silently; name both sources and ask staff to resolve.
FM12 Missing supplied evidence	Allow provisional intake, list missing source and owned next step; no invented claim.
FM13 Forged citation	Verify citation ID/version/quote against frozen authorized view, never only syntax.

Risk scores are proposed estimates. They are not measured odds or risk acceptance.

Components and contracts

This inventory is a summary. Exact typed inputs, outputs and permission rules are in architecture.md and the build milestone files.

Component / group	Responsibility	Task IDs
C1 / Code: validation	Check case envelope	T1
C2 / Code: tools	Build trusted evidence view	T2
C3 / Agents (LLM)	Draft bounded recommendation	T3
C4 / Code: validation	Check claims and output proof	T3, T4
C5 / Code: human triggers	Prepare human handoff	T4

Shared control boundary

Code checks identity, money, dates and limits. A model may suggest a choice. It cannot authorise or perform a business write.

Architecture and model budget

F1 - Workflow choice

Workflow choice

Why: Bound the work and make the result reviewable.

F2 - Permissions and data

Permissions and data

Why: Bound the work and make the result reviewable.

F3 - Storage and deployment options

Storage and deployment options

Why: Bound the work and make the result reviewable.

Model cost

Illustrative model cost is USD 0.00720 per case. Unknown prices are not free service. Confirm rates, traffic, staff and tools before launch.

Full call counts, token budgets, rejected choices and costs: design.json, architecture.md and assumptions.md.

Evaluation coverage matrix

Risk / task	EV1	EV2	EV3	EV4	EV5	EV6	EV7	EV8	EV9	EV10	EV14	EV15
FM1	-	X	-	-	-	-	-	-	-	-	-	-
FM2	-	-	X	-	-	-	-	-	-	-	-	-
FM3	-	-	-	X	-	-	-	-	-	-	-	-
FM4	-	-	-	-	X	-	-	-	-	-	-	-
FM5	-	-	-	-	-	X	-	-	-	-	-	-
FM6	-	-	-	-	-	-	X	-	-	-	-	-
FM7	-	-	-	-	-	-	-	X	-	-	-	-
FM8	-	-	-	-	-	-	-	-	X	-	-	-
FM9	-	-	-	-	-	-	-	-	-	X	-	-
FM10	-	-	-	-	-	-	-	-	-	-	X	-
FM11	-	-	-	-	-	-	-	-	-	-	-	X
FM12	-	-	-	-	-	-	-	-	-	-	-	-
FM13	-	-	-	-	-	-	-	-	-	-	-	-
T1	X	X	X	X	X	-	X	-	X	-	X	-
T2	X	X	X	X	-	-	-	-	-	-	-	X
T4	X	X	X	X	X	X	X	X	X	X	X	X

Evaluation coverage matrix - continued

Risk / task	EV16	EV17
FM1	-	-
FM2	-	-
FM3	-	-
FM4	-	-
FM5	-	-
FM6	-	-
FM7	-	-
FM8	-	-
FM9	-	-
FM10	-	-
FM11	-	-
FM12	X	-
FM13	-	X
T1	X	-
T2	X	-
T4	X	X

X means the test is linked to that risk or task. It does not mean the test passed. Complete fixtures and assertions: eval-set.json.

Evaluation plan

Test	Layer	Purpose
EV1	golden	Useful bounded packet
EV2	golden	Currency differs
EV3	golden	Ambiguous OCR digit
EV4	golden	Wrong supplier PO
EV5	golden	Revoked source access
EV6	golden	Model output unavailable
EV7	golden	Stale concurrent replacement
EV8	golden	Missing receipt
EV9	golden	Cancellation
EV10	golden	Overdue human review
EV11	offline	Held-out staff usefulness and evidence audit
EV12	online	Review-only operating monitor
EV13	release	Release gate for all specified branches
EV14	golden	Exact duplicate request
EV15	golden	Conflicting supplied evidence
EV16	golden	Missing supplied evidence
EV17	golden	Forged citation

These are test plans. No business controls, provider quality or user outcomes were measured. See eval-set.json for fixtures and pass rules.

Measurement and rollout

R1 - Business measure

Business measure

Why: Bound the work and make the result reviewable.

R2 - Shadow on historical cases

Shadow on historical cases

Why: Bound the work and make the result reviewable.

R3 - Assist-only pilot

Assist-only pilot

Why: Bound the work and make the result reviewable.

R4 - Measure review effort

Measure review effort

Why: Bound the work and make the result reviewable.

R5 - Accountable owner and stop plan

Accountable owner and stop plan

Why: Bound the work and make the result reviewable.

Build handoff and review

Start with build/00-overview.md. Load one milestone file at a time. Exact source rules remain in design.json; do not treat design approval as live write permission.

X1: Build a review-only tool

Review dimensions - evidence in review.json

Dimension	Score	Dimension	Score
D1	3/4	D7	3/4
D2	3/4	D8	3/4
D3	3/4	D9	3/4
D4	3/4	D10	3/4
D5	3/4	D11	3/4
D6	3/4	D12	3/4

I1 major resolved: W1-W5 decision fields were authored as strings rather than booleans; structural validation fails. Repair type without removing predicates from description.

I2 major resolved: F1-F3 alternatives were authored as strings rather than arrays; structural validation fails.

I3 major resolved: Exceptional paths reach C5 without a C4.checked value. Declare a typed union handoff envelope carrying exception evidence; do not require an absent successful artifact.

I4 major resolved: Supplement domain cases with exact duplicate, evidence conflict, missing source and forged citation cases; release must include these actual IDs.

Glossary

canonical: agreed

discriminated: typed

payload binding: approved details

digest: fingerprint

idempotency: duplicate-write protection

reciprocal: two-way

terminal disposition: final outcome

calc_digest: a fingerprint of the approved amount

After delivering DESIGN.pdf, ask the user: Would you also like a visual HTML version in the same ten-stage structure as the app? Generate HTML only after opt-in, unless already requested.

Full assumptions, owners and evidence: assumptions.md. Review method and failed rounds are preserved in review.json and review-round-*.json.